

Introduction:

Rival Robotics is a competition where teams build 20-pound (9-kilogram) robots with no restrictions on materials, electronics, and mechanical subsystems. This flexibility allows teams to create anything from extremely simple robots to highly complex machines. Each year, a new game manual is released at the beginning of the season, defining the challenge to which each team's robot must be designed. This extended timeline allows teams to iterate and improve their robots between competitions significantly.

2024 Game Overview:

The 2024 Rival Robotics game was inspired by the video game Rocket League. Teams competed to maneuver a soccer ball into goals located within a small, enclosed arena. Matches were played 2-vs-2, with each match lasting two minutes. Robots were allowed to extend slightly beyond their frame perimeter, adding an extra layer of strategic depth to the gameplay.

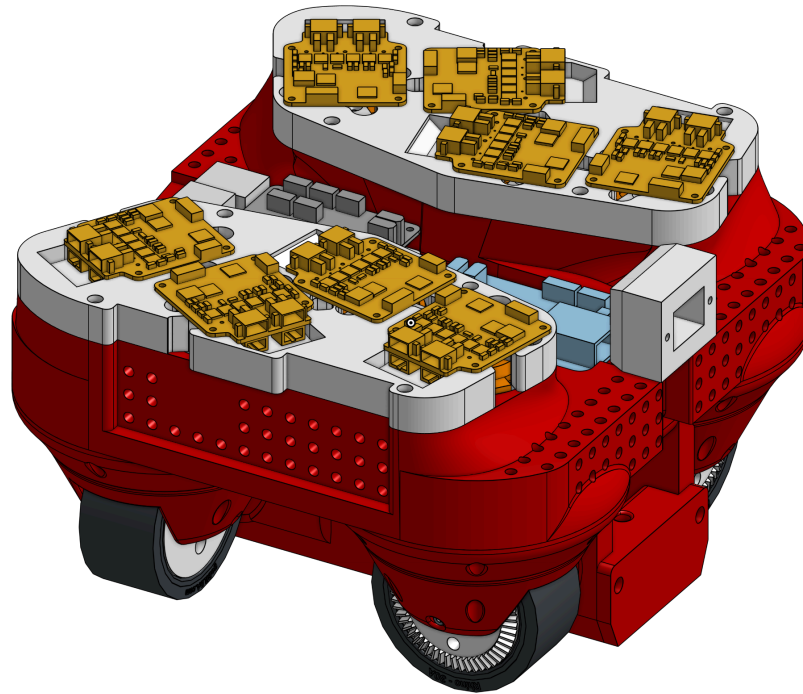
Our Strategy Choice

Our main goal was to effectively maneuver around the field quickly and efficiently. We chose a swerve drive base for several reasons:

- Speed:
 - We wanted the robot to move quickly around the arena without wasting time on orientation changes — being able to drive in any direction at full speed was a key advantage.
- Traction:
 - With four wheels gripping the ground at all times, we had consistent traction in every direction, whenever we needed it.
- Defense:
 - Our robot couldn't easily be pushed or T-boned by opposing robots. It could always escape quickly and effectively.
- Maneuverability:
 - This is essentially the combination of speed, traction, and defense: our robot could go wherever we wanted, whenever we wanted.

Originally, we planned to develop a game piece effector to help move the soccer ball, inspired by the robot Impact. However, we ultimately decided it wasn't necessary for success. With limited resources and time, we focused on perfecting our drivetrain instead. In the end, we avoided scope creep!

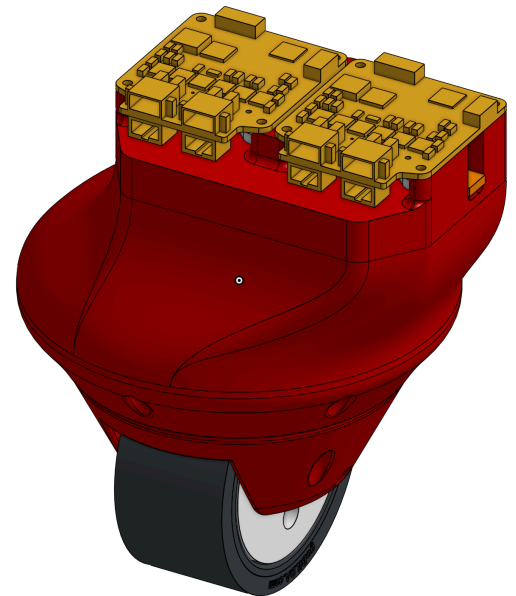
How Dash Is Made:

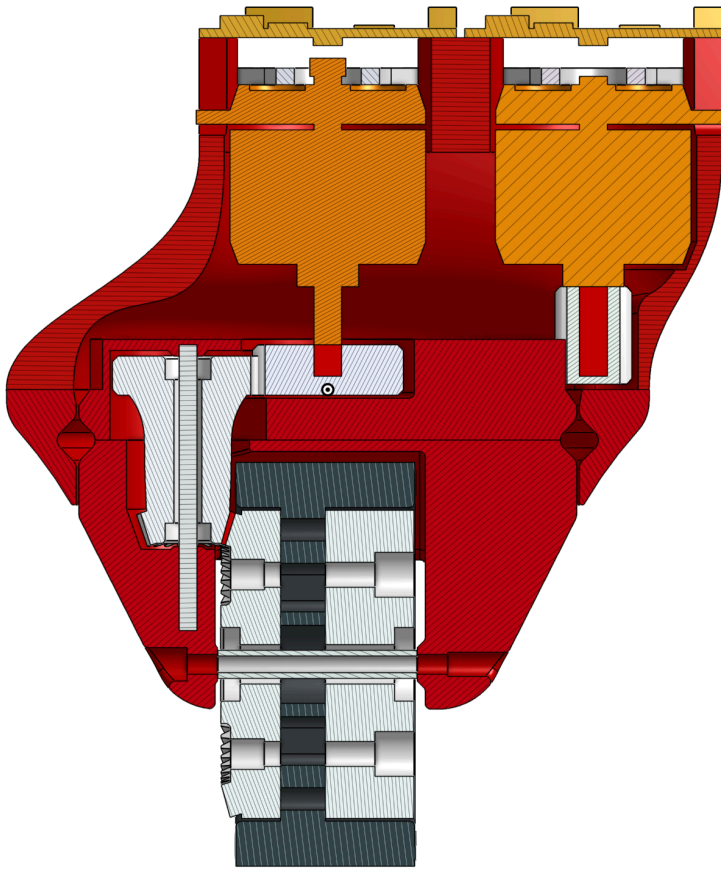


Dash, our robot, is almost entirely 3D printed excluding electronics and hardware. This introduced a unique set of design constraints. Each part had to be robust, match the tolerances achievable by an X1C printer, and be easy to replace in case of failure. This meant that some parts—especially those in the swerve modules—were larger than you’d typically find in COTS (Commercial Off-The-Shelf) FRC modules.

Dash uses four swerve modules arranged in a square for maximum stability. The robot runs on eight brushless motors, each paired with a Moteus CAN-based ESC. These ESCs include built-in encoders, allowing for precise position and velocity control.

Let's break down one of the modules of Dash so that we can see its inner workings of it. Each module is made up of a few different components. The first one is the wheel. We chose to use goBILDA Rhino wheels. These wheels are very grippy and wide, which makes them the perfect candidate when needing a lot of traction. There are two 3536 brushless motors per module: one in charge of steering, and the other driving. Let's take a look at a cross-section of the module and break down some different design choices.

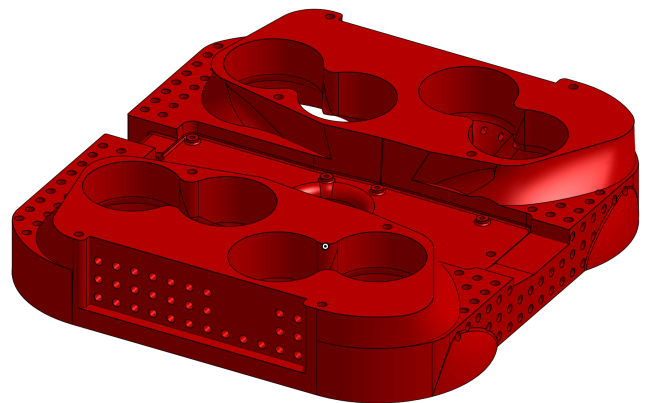


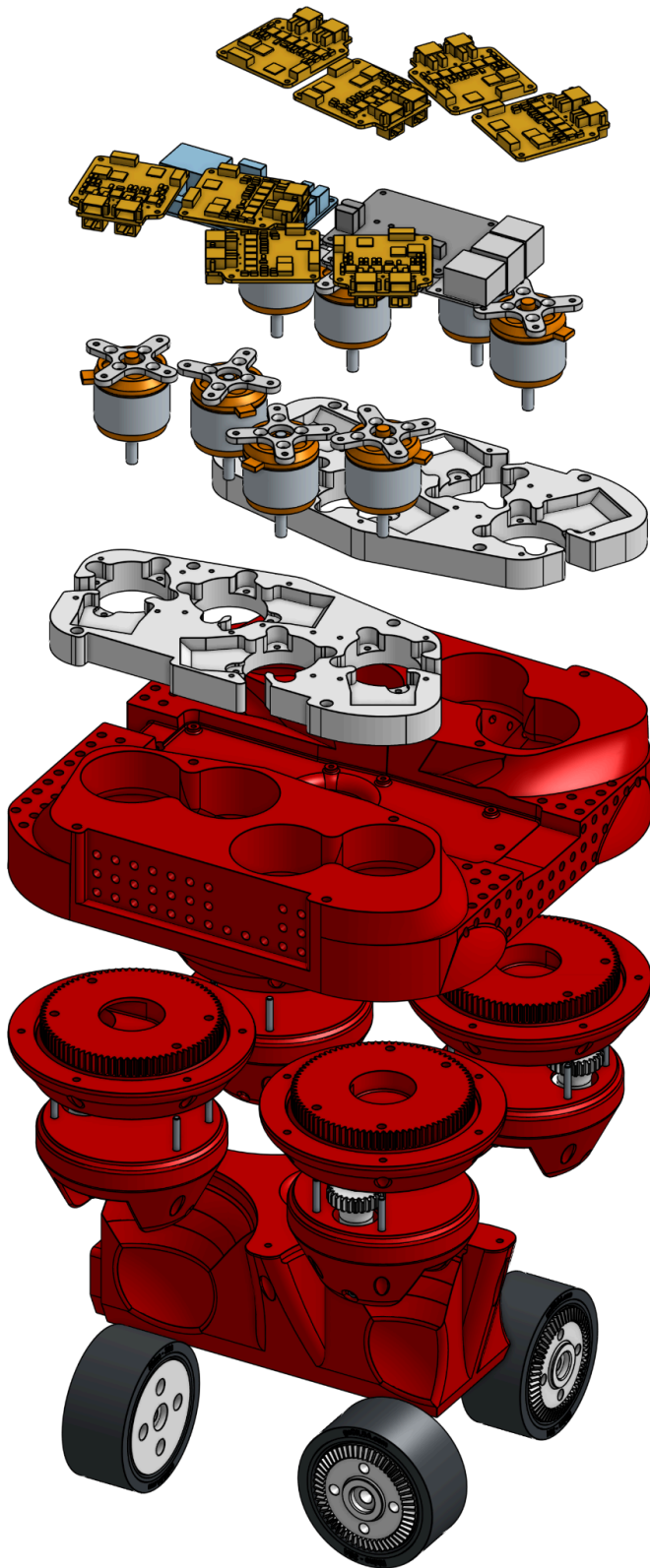


The center motor is the drive motor, which powers a central gear that is the basis of drive movement. That then goes to a 3D printed through-gear which takes the power from the motor and transitions it to a bevel gear that connects with the wheel. There is a small steel shaft that goes through this gear to provide stability and support with the power transfer. It then moves to the wheel where we have made a custom hub that turns the wheel into a dead axle, and it gets powered off of a bevel gear on one side of it.

The motor to the right is the steering motor, which is in charge of the heading of the module. It uses a ring gear on the outside of the central module to be able to change the direction. The bearing is made out of plastic airsoft BBs due to their availability and low price. This is different from the standard of using a large roller bearing, but we already had the BBs on hand.

We changed from having an individual body for each of the modules to having one larger body that fits all of the modules inside of it. This change decreased complexity and assembly time but also helped increase the rigidity of the chassis, which made it much stronger and more resistant to any moments that might be placed on it. Swerve needs to have a rigid and sturdy mounting to make it so that the encoder positions stay accurate and so that you can apply all of the traction and benefits that you get from swerve.





Let's break down some different components of the assembly of the robot. It continues with the idea of segmenting things and making them both easy to 3D print and to assemble, focusing on big and robust parts that serve multiple purposes.

The motors are all mounted on a metal mount. The reason for this is to resist any moments that might be exerted onto the mount when in motion. Due to these being small and lightweight, it was not an issue to integrate them into the design of the robot.

The mounts then fasten onto the white plates. These plates' purpose is to hold both the motors and the Moteus controllers in place. They are mirrored parts which have the motors offset into the piece to get the exact offset needed from the controller. The controller then mounts on top of the motor using heat-set inserts. We utilize these inserts extensively for their ease of installation and their strength versus plastic threads.

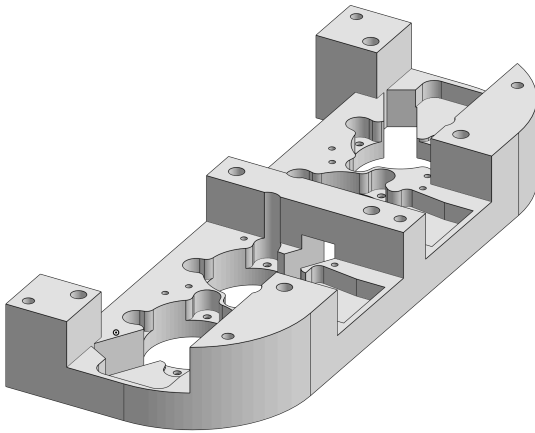
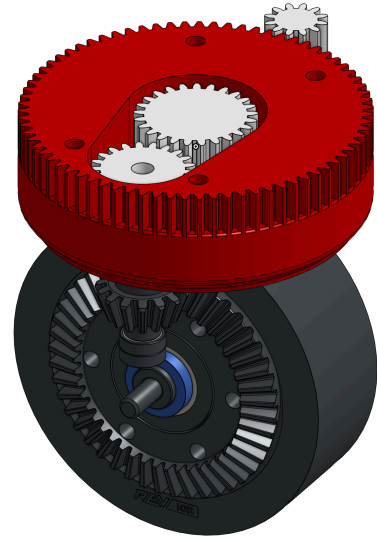
The main plate is made in one large piece. Fun fact: it is the maximum size that you can fit onto one build plate of an X1C. All of the holes on the outside accommodate heat-set inserts for mounting things on the outside of the robot with ease.

The module is made up of two main parts: the top piece and then the forks section. These two sections are bolted together to trap the BBs to allow for a seamless ball bearing.

The bottom part is the battery compartment, which is one big box that houses the battery on the bottom section of the robot. This is to reduce the center of gravity and make the robot much harder to tip. With a large majority of the parts being 3D printed, it allows for the CG to easily be changed with components like the battery.

Changes Throughout the Season:

There are three main changes to Dash while competing. The first of those changes was switching to a metal bevel gear set for our swerve modules. The plastic gears worked but wear started to become an issue quickly. To replace worn bevels, we had to fully disassemble the modules, making it unfeasible to continue using 3D-printed gears. We switched to using a majority of the power transfer system of a REV Robotics MAXSwerve module for the lower part of the module, including the gearing and wheel. This change made it so that throughout the competition we would not see wear on the gears as much.



The second part that we iterated on was the motor mounts. The changes here were not out of necessity but just out of convenience—changing the modules to be oriented in a fashion that made wiring and accessibility of the ports easier.

The last major change was adding a bottom bumper and changing the battery compartment. We added TPU bumpers to the bottom of the robot to both lower the CG and make it harder to tip due to having a wider surface area. We also integrated the battery compartment in an easier and nicer-looking method. This also helped increase the overall rigidity of the robot.

